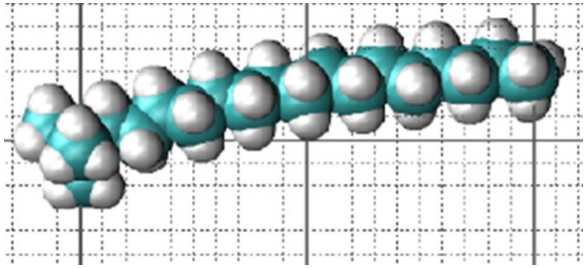
Molecular Dynamics: the idea

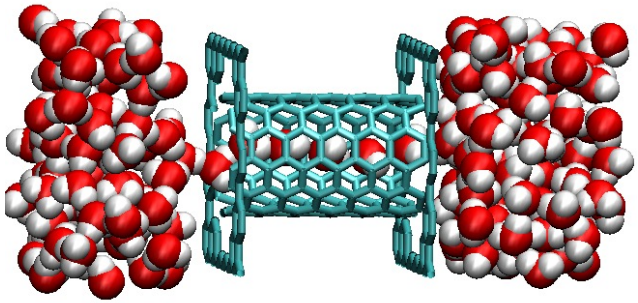
ICMAB
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- We know that matter is made by **atoms**





- We know the underlying physical/chemical principles for atomic/molecular energies, interactions and motion

$$[-\frac{\hbar^2}{2m}\nabla^2 + V]\Psi = E\Psi$$

$$p(E) \propto \exp(-E/k_{\mathrm{B}}T)$$

$$\vec{a}(t) = \frac{\vec{F}(t)}{m}$$

It has to be possible to predict the properties and behaviour of matter combining all this knowledge

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Studies in Molecular Dynamics. I. General Method*

B. J. ALDER AND T. E. WAINWRIGHT

Lawrence Radiation Laboratory, University of California, Livermore, California

(Received February 19, 1959)

A method is outlined by which it is possible to calculate exactly the behavior of several hundred interacting classical particles. The study of this many-body problem is carried out by an electronic computer which solves numerically the simultaneous equations of motion. The limitations of this numerical scheme are enumerated and the important steps in making the program efficient on the computers are indicated. The applicability of this method to the solution of many problems in both equilibrium and nonequilibrium statistical mechanics is discussed.

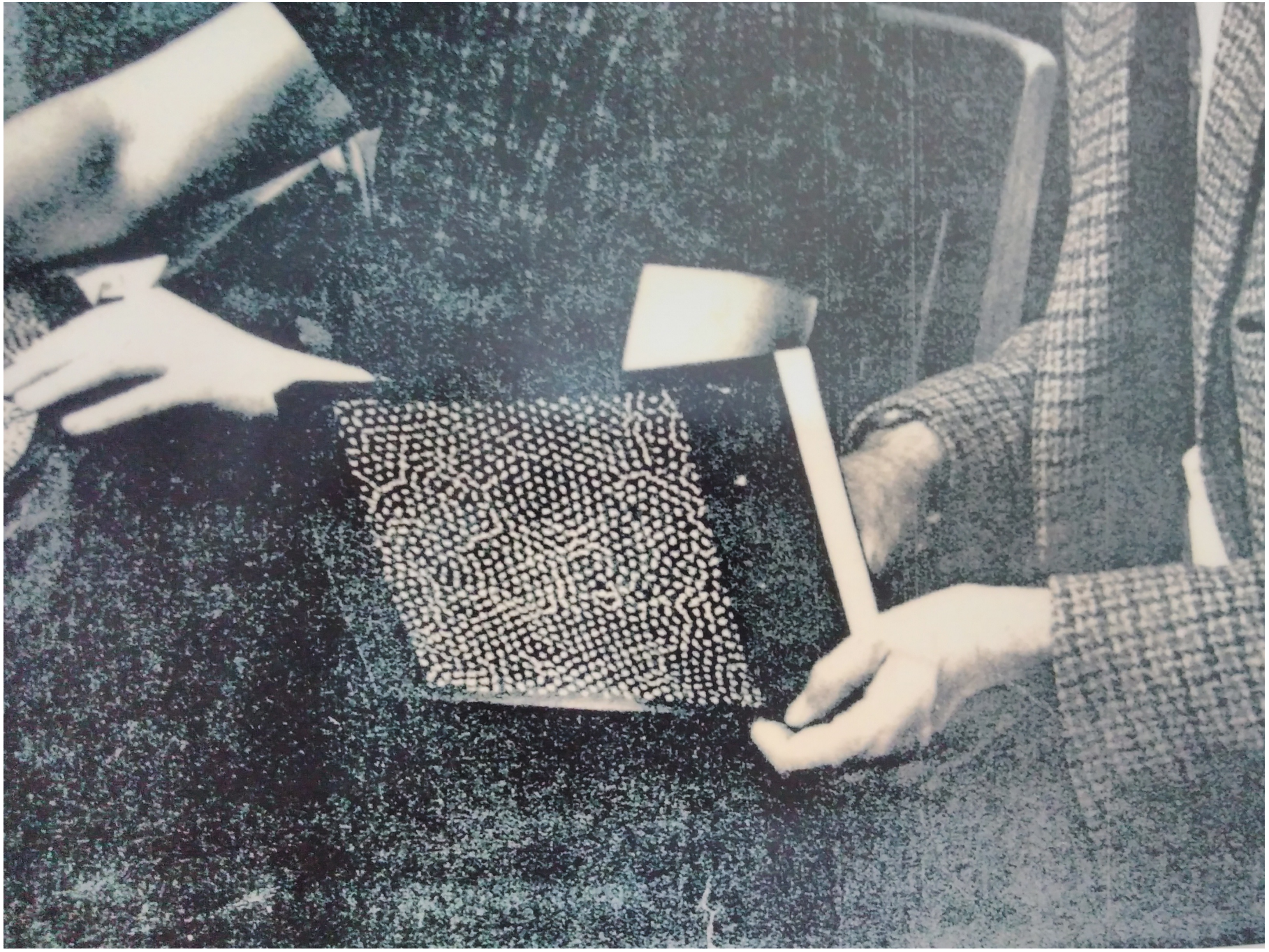
https://www.youtube.com/playlist?list=PLWhDHF4i-Is2fkngQOvfO1lGhG3P5ebAa



Berni Alder (standing) with Mary Ann Mansigh (coder) and Tom E. Wainwright (physicist) in his office, 1962, "looking at oscillograph traces from the computer generated motion of particles. The pictures show the coexistence of cristallites and melt for two dimensional system of elastic disks". Courtesy AIP, Emilio Segré's visual Archives



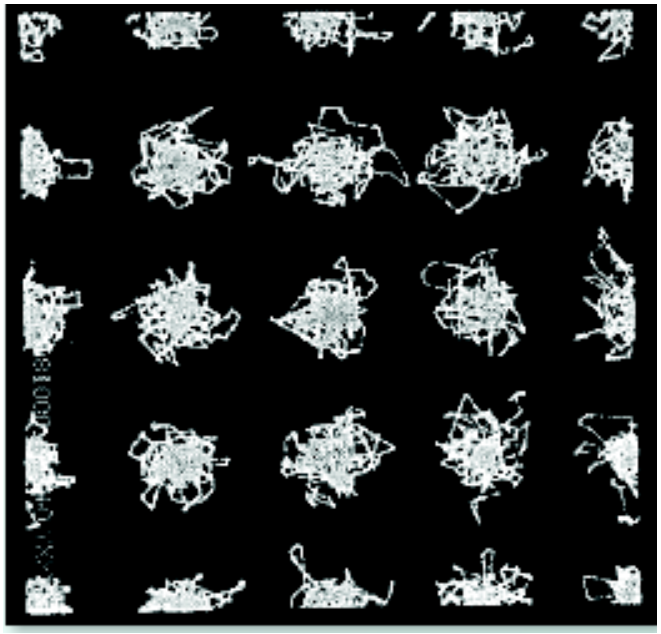
Berni Alder (standing) with Mary Ann Mansigh (coder) and Tom E. Wainwright (physicist) in his office, 1962, "looking at oscillograph traces from the computer generated motion of particles. The pictures show the coexistence of cristallites and melt for two dimensional system of elastic disks". Courtesy AIP, Emilio Segré's visual Archives



First paper in MD:

Alder and Wainwright, “Studies in Molecular Dynamics”,
J. Chem. Phys. 31, 459 (1959)

FIG. 2. The traces of 32 hard sphere particles in the periodic boundary conditions in the solid phase for about 3000 collisions.



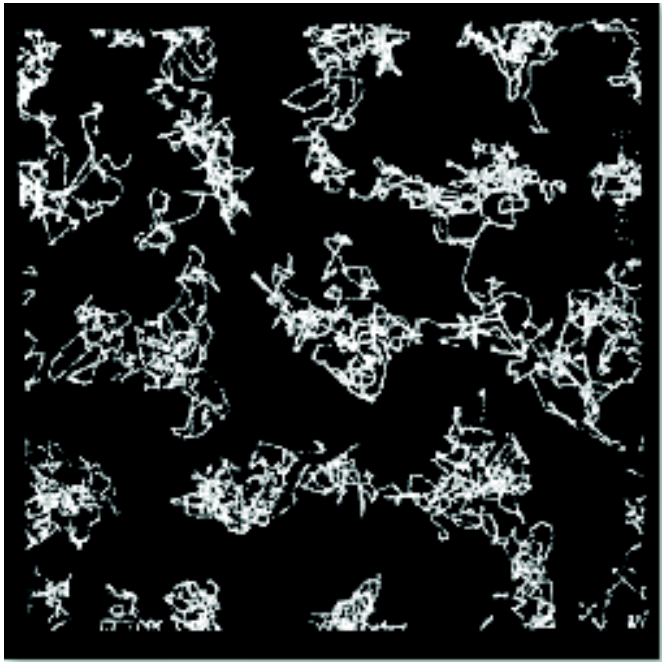
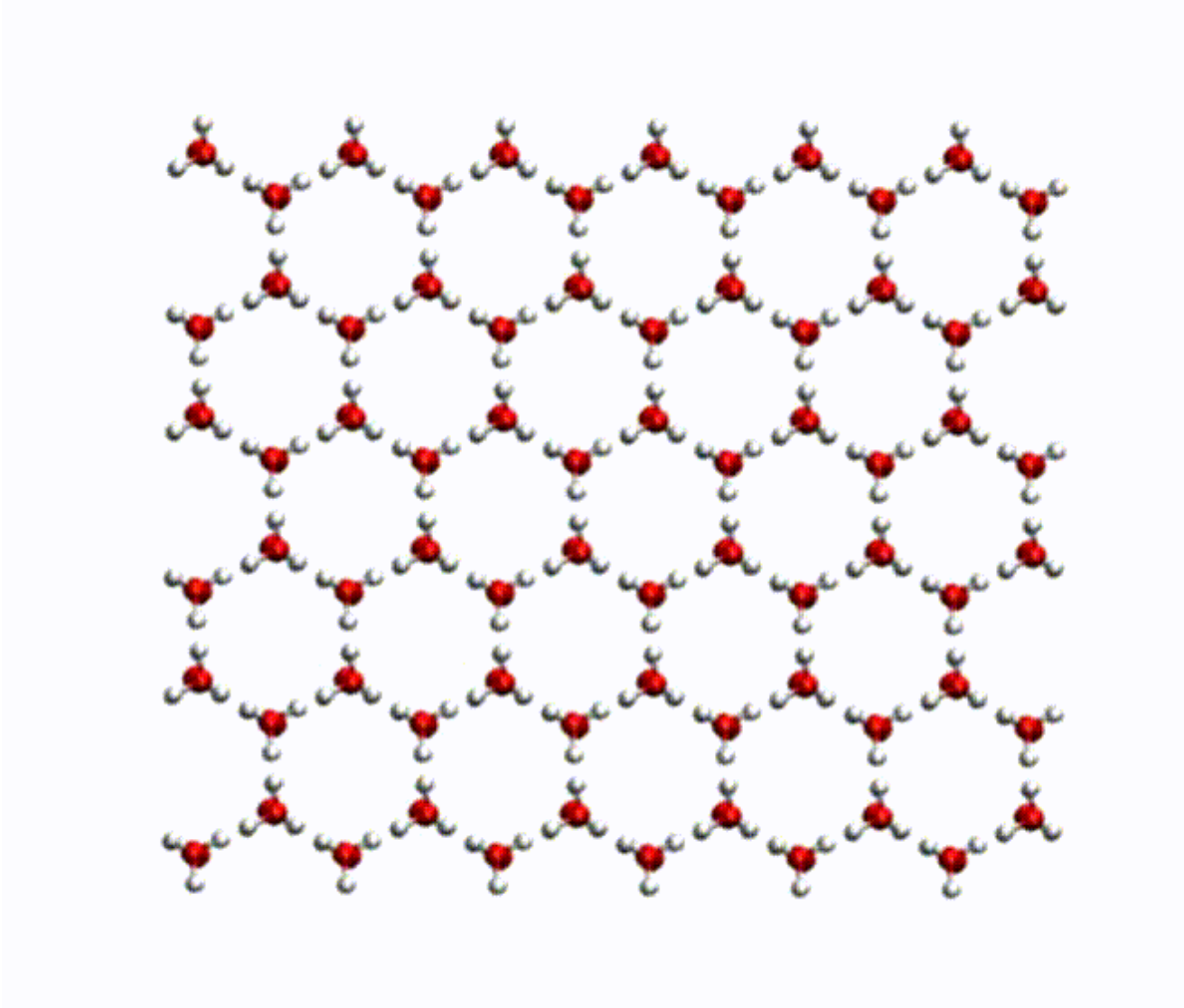


FIG. 3. The traces of the same system as in Fig. 2 after it has transformed to the fluid phase also for 3000 collisions.

The same exemple but water instead of hard spheres



https://youtu.be/_Upr-uSLvGM?si=MR5bnG-ujZhMKK7C

Online/downloadable demonstration of SoftMatter **Molecular Dynamics (MD) simulations:**

https://softmatterdemos.org/SADemo.php

OBJECTIVE: Show that the interplay between interactions and thermal effects gives rise to complex behaviour including self-assembly

- The particles in the simulation are considered as classical entities, obeying Newton's Laws of motion.

$$\vec{F}_i = m_i \vec{a}_i \qquad \vec{a}_i = \frac{d\vec{v}_i}{dt} \qquad \vec{v}_i = \frac{d\vec{r}_i}{dt}$$

- The velocities of the particles are thermalized, always **following Maxwell-Boltzmann distribution**

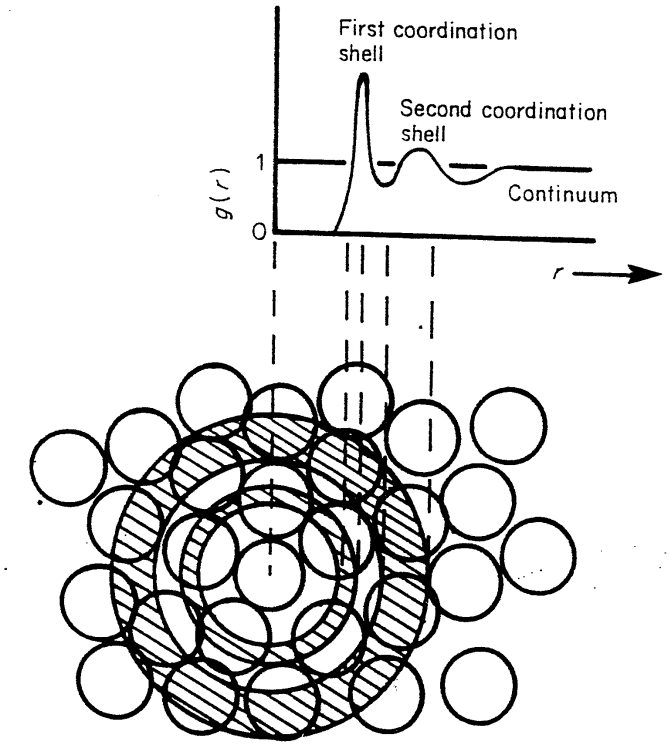
- The **forces between the particles** can be selected to correspond to different interactions that correspond to **nanoparticles or colloidal particles in solution.**

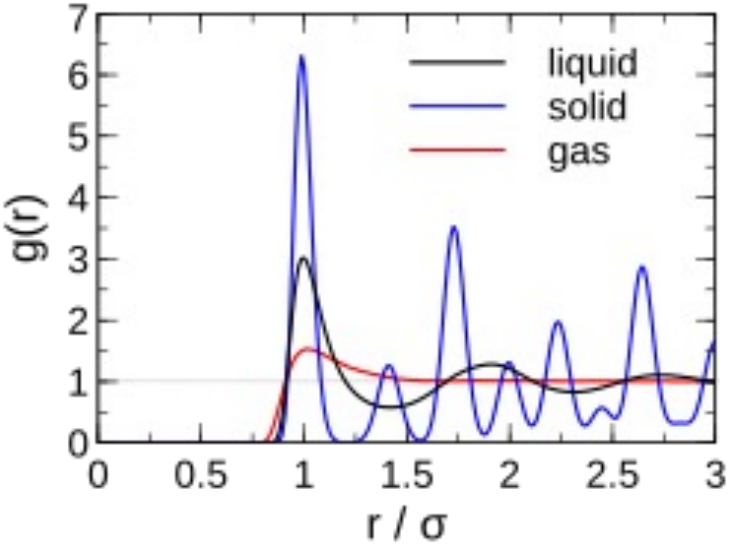
Output quantities from these interactive MD simulations:

- Potential Energy
- **Temperature** and **Pressure** computed by equipartition and virial theorems

$$E_{kin}(t) = \frac{1}{2} N_{\mathrm{fr}} k_B T_{inst}(t) \qquad P = \rho k_B T + \frac{1}{3V} \left\langle \sum_{i<j} \vec{r}_{ij} \vec{F}(\vec{r}_{ij}) \right\rangle$$

- Radial distribution (correlation) function g(r): local density divided by the average density (g=1 for an ideal gas)





Some Soft Matter Examples

See our YouTube channel

https://www.youtube.com/channel/UCkrTDh1VQw06o0dzPcF2NgA

